

OSSP (Operating Systems And Systems Programming) - 25CS2104E

Roll No. : 2520030029

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SEC-8

SKILL-2:

Aim: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer,
Support Multi-Character Commands, Test User Interaction

```
siddinapujith07@DESKTOP-FDHCAIT:~$ cd ~
siddinapujith07@DESKTOP-FDHCAIT:~$ mkdir LinuxShellProjectS2
siddinapujith07@DESKTOP-FDHCAIT:~$ cd LinuxShellProjectS2
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ mkdir src include
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ touch Makefile
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ touch README.md
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ touch src/main.c
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ ls -R
.:
Makefile  README.md  build  docs  include  src  tests
./build:
./docs:
./include:
./src:
main.c

./tests:
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ nano Makefile
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ nano src/main.c
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ make
gcc -Wall src/main.c -o shell
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ ./shell
myshell>
You entered:
myshell> hello
You entered: hello
myshell> exit
Exiting Shell...
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ char input[100]
;
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'car' from deb ucommon-utils (7.0.1-1)
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ while (1)
>
> exit
>
> exit
> printf("myshell> ");
-bash: syntax error near unexpected token `"myshell> "'
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ printf("myshell> ");
-bash: syntax error near unexpected token `"myshell> "'
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ printf("myshell> ");
-bash: syntax error near unexpected token `"myshell> "'
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProjectS2$ fgets(input, sizeof(input), stdin);
-bash: syntax error near unexpected token `input,'
```

Code:

```
</> Makefile

CC = gcc
CFLAGS = -Wall
TARGET = shell

all:

    gcc -Wall src/main.c -o shell

clean:

    rm -f shell
```

```
#include <stdio.h>
#include <string.h>

int main()
{
    char input[100];

    while (1)
    {
        printf("myshell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

        if (strcmp(input, "exit\n") == 0)
        {
            printf("Exiting Shell...\n");
            break;
        }

        printf("You entered: %s", input);
    }

    return 0;
}
```